11) Zeolite is sodium aluminium silicate.

$(NaAlSi_2O_6 \cdot H_2O)$

Option (a)

12) (a) 1 mL of H_2O_2 will give 100ml O_2 at STP.

13) $Cr_2O_7^{2-} + 2H^+ + 4H_2O_2 \rightarrow 2CrO(O_2)_2 + 5H_2O$

Option (c)

14) $2MnO_4^- + 5H_2O_2(aq) + 6H^+ \rightarrow 2Mn^{2+} + 5O_2 + 8H_2O$

Option (c) 5/2 moles

15) Volume strength of hydrogen peroxide = Normality of hydrogen peroxide × 5.6

$= 1.5 \times 5.6$

$= 8.4$

Option (d)

$$\underset{(2\times 34g)}{2H_2O_2} \rightarrow 2H_2O + \underset{(22.4\ litres)}{O_2}\uparrow$$

Volume strength of Hydrogen peroxide

$$= \frac{\text{Normality} \times \text{Equivalent weight of } H_2O_2 \times 22.4}{68}$$

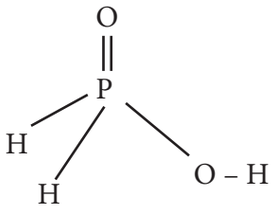
$$= \text{Normality} \times \left(\frac{17\times 22.4}{68}\right)$$

Volume strength of hydrogen peroxide
= Normality × 5.6

16) sp^3 and sp^3

Option (d)

17) Hypophosphorus acid on reaction with D_2O, only one hydrogen is replaced by deuterium and hence it is mono basic

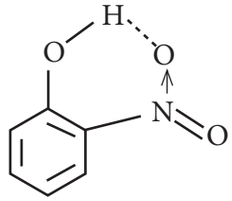


Option (c) monobasic acid

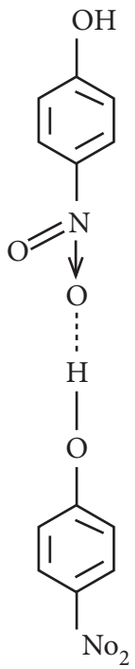
18) (a) tetrahedrally surrounded by 4 hydrogen atoms (refer 4.6 (a) Structure of ice)

19)

o-nitro phenol



p - nitro phenol



Option (b)

20) Heavy water is used as moderator as well as coolant in nuclear reactions.

Option (c)

21) Water is a amphoteric oxide.

Option (c)

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